Exercise 7

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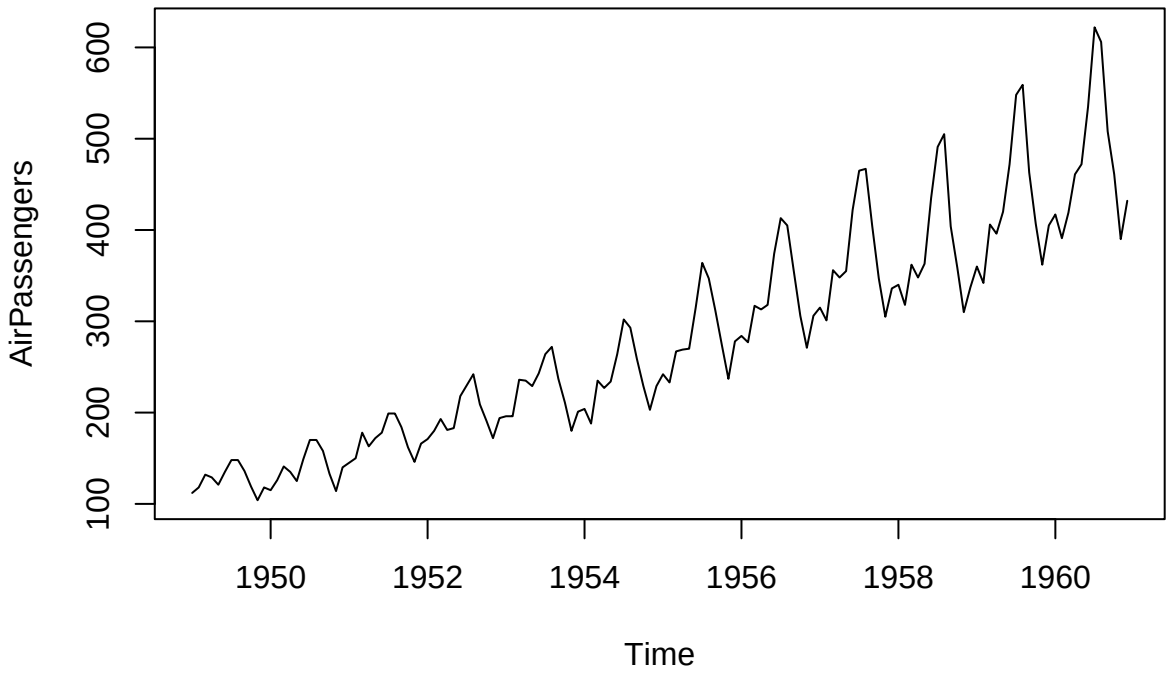
2026-03-28

a)

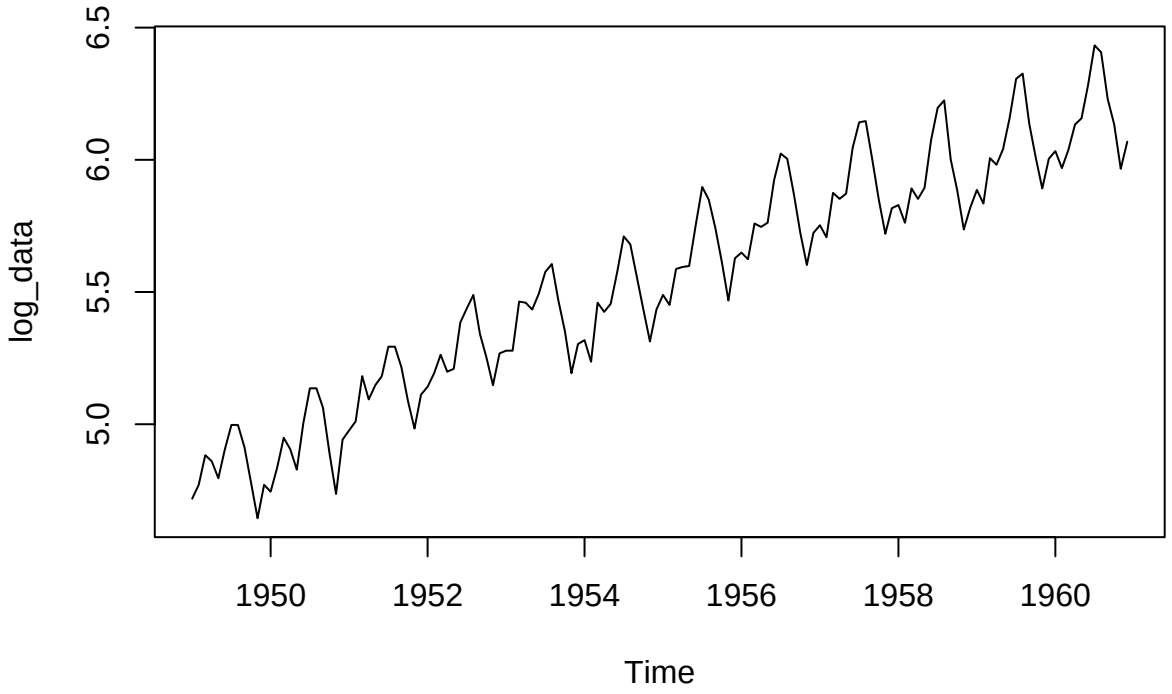
```
data("airpass")
```

```
## Warning in data("airpass"): data set 'airpass' not found
```

```
plot(AirPassengers)
```



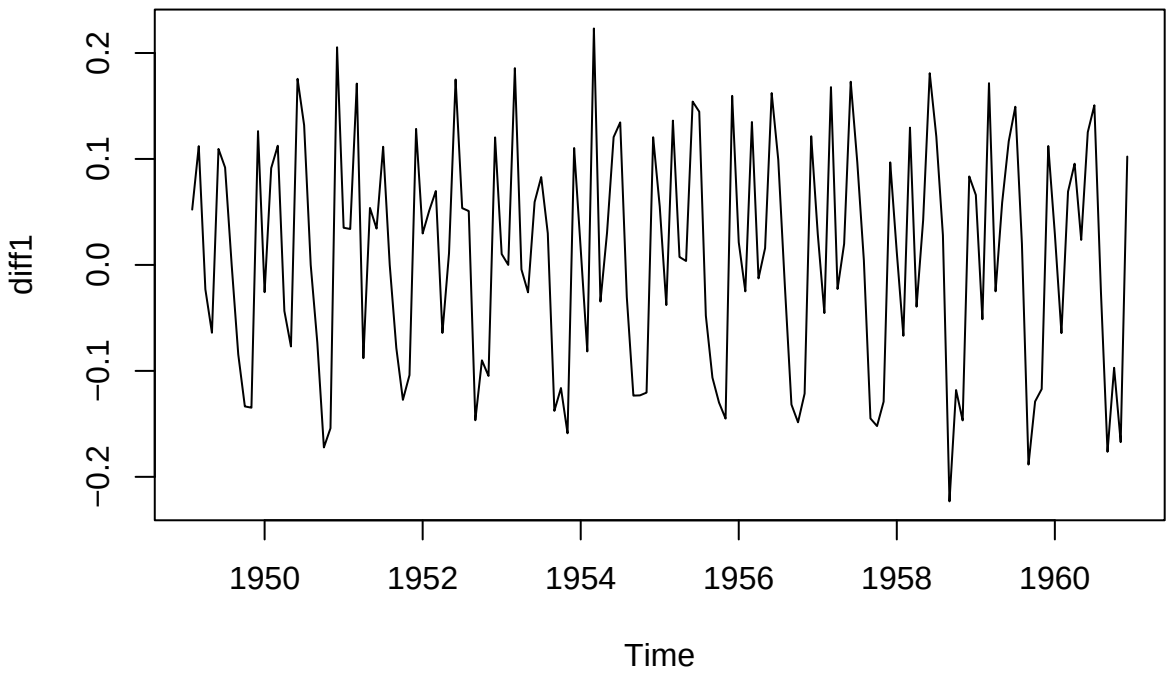

```
log_data = log(AirPassengers)
plot(log_data)
```



Taking the log is an appropriate transformation because it the reduce the scale of the data without any information loss.

b)

```
diff1 = diff(log_data, 1)
plot(diff1)
```

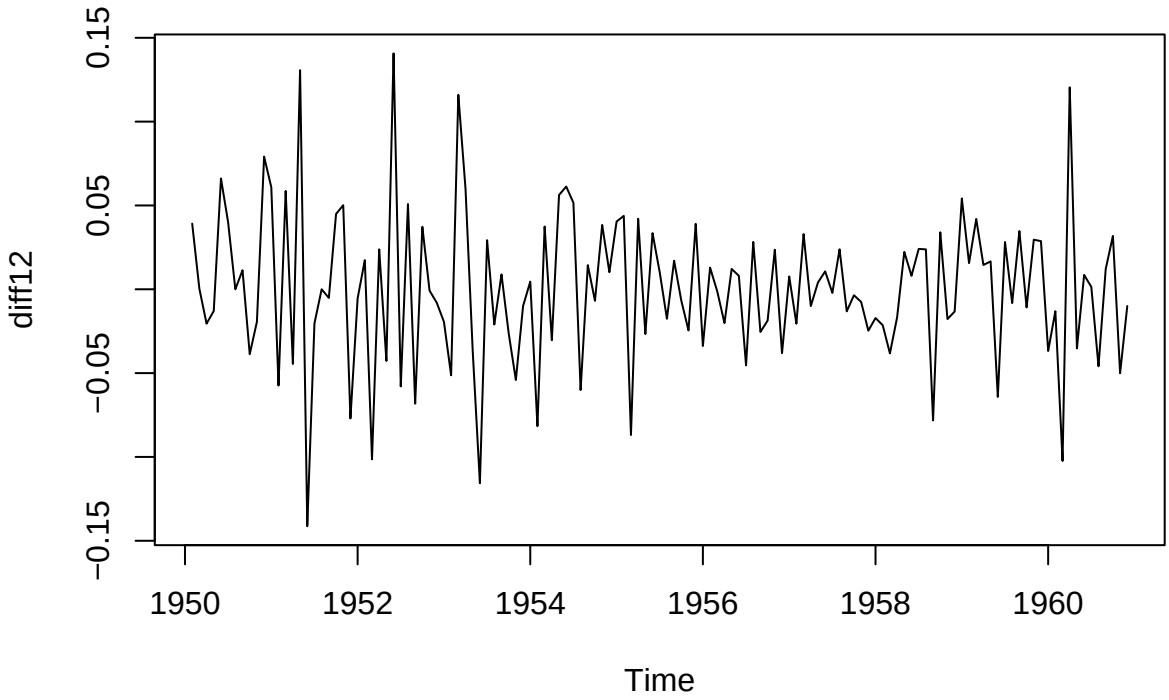


Taking the first difference of the logged data yields a time serie without any trend and probably stationary. Moreover

the data point hang together.

c)

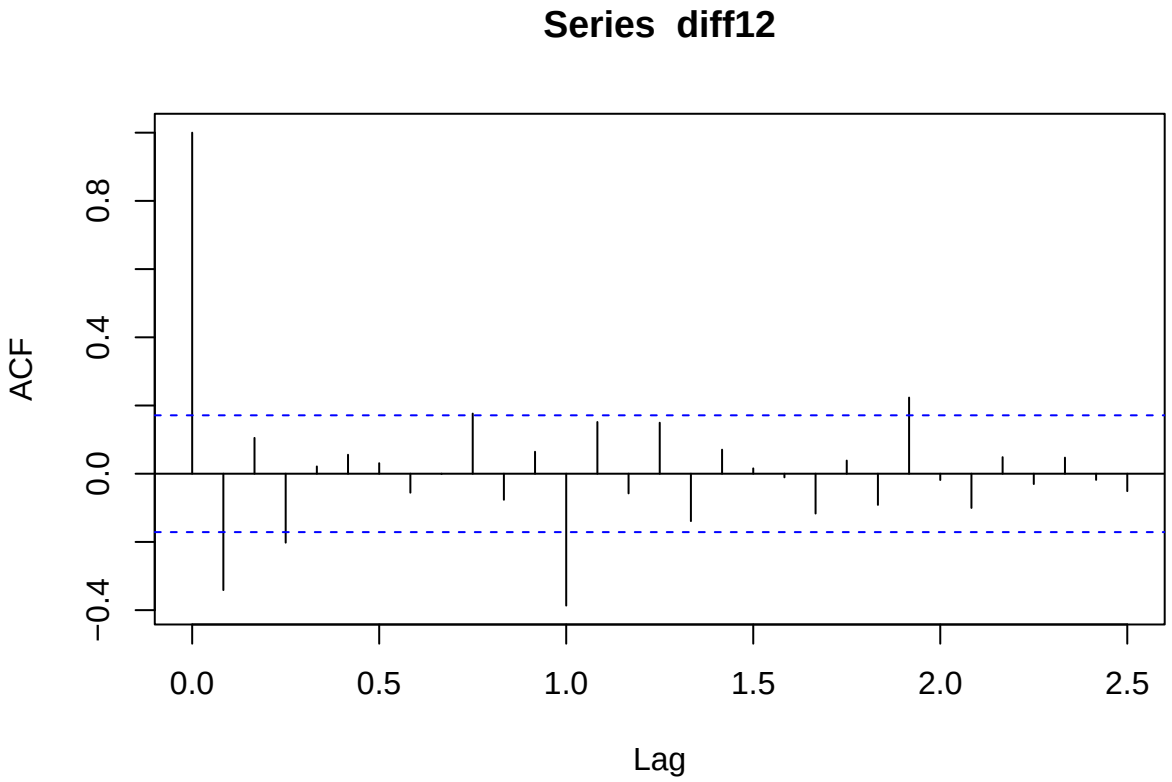
```
diff12 = diff(diff1, 12)
plot(diff12)
```



The resulting time serie seems to be still stationary without any appearing trend and the data point hang together less oftenly.

d)

```
acf(diff12, lag.max=30)
```



The ACF shows a stationary series after seasonal differencing, with a strong MA(1) behavior at lag 1 and a clear seasonal MA effect at lag 12

e)

```
library(forecast)
```

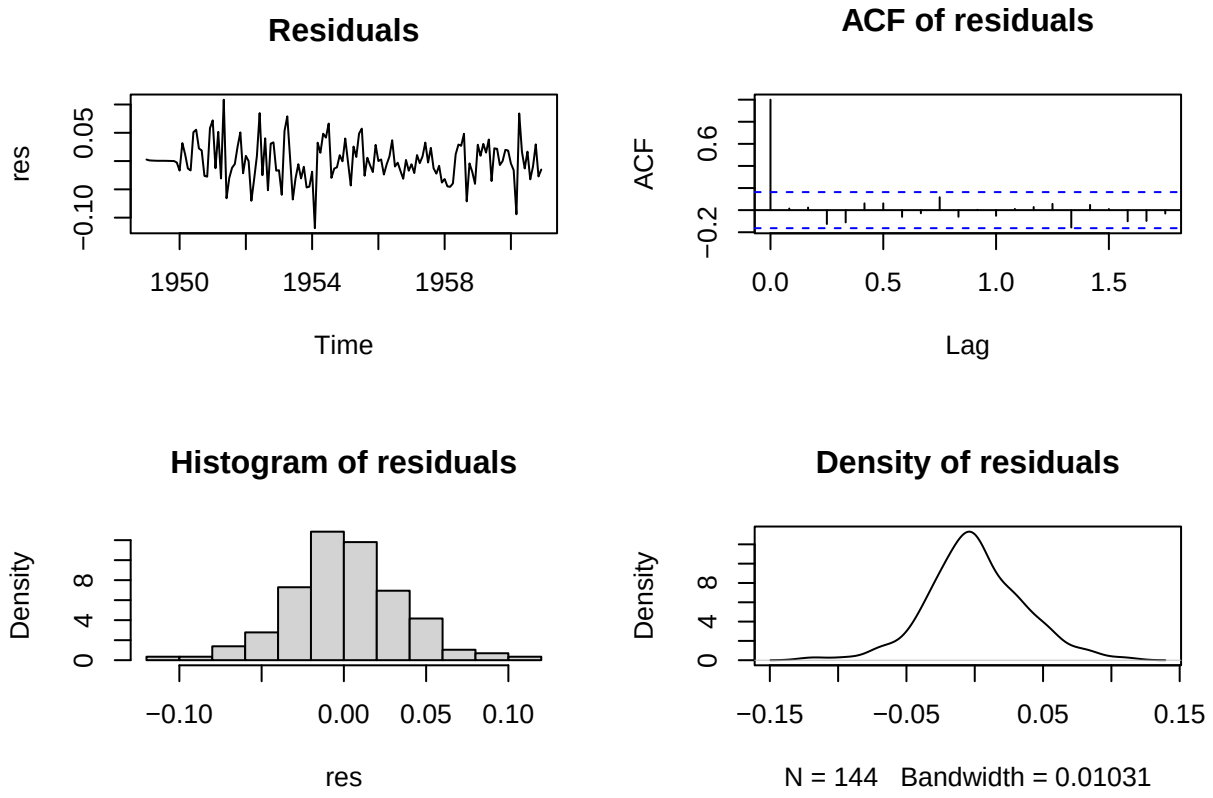
```
## Warning: package 'forecast' was built under R version 4.4.3
```

```
# SARIMA (0,1,1)(0,1,1)[12]
fit <- Arima(log_data
  ,
  order = c(0, 1, 1),
  seasonal = list(order = c(0, 1, 1), period = 12)
)
```

f)

```
res <- residuals(fit)
par(mfrow = c(2,2))
plot(res, main = "Residuals")
```

```r
acf(res, main = "ACF of residuals")
hist(res, main = "Histogram of residuals", freq = FALSE)
plot(density(res), main = "Density of residuals")
```



Based on the plots, the residuals behave like white noise.

```r
Box.test(res, lag = 20, type = "Ljung-Box")
```

```
##
##  Box-Ljung test
##
## data:  res
## X-squared = 17.688, df = 20, p-value = 0.6079
```

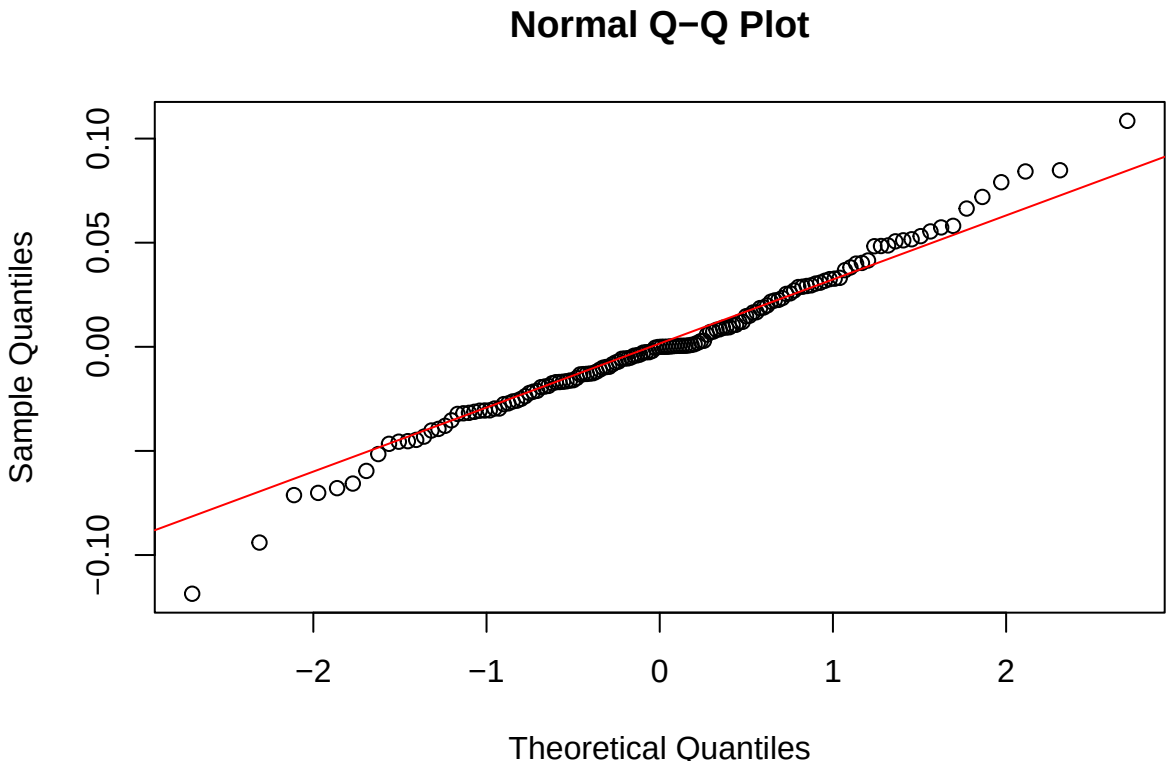
pvalue $= 0.6 > 0.05$ so there is no remaining autocorrelation and the residuals behave like white noise.

```r
shapiro.test(res)
```

```
##
##  Shapiro-Wilk normality test
##
## data:  res
## W = 0.98637, p-value = 0.1674
```

pvalue $= 0.16 > 0.05$ so we have no evidence against the normality of the residuals.

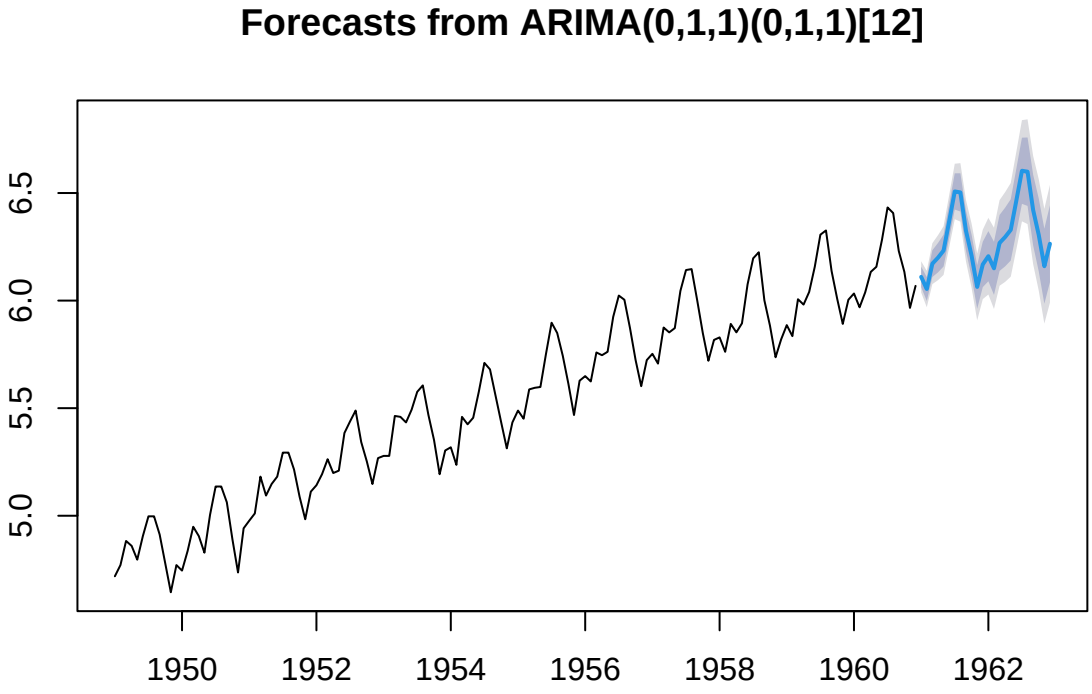
```r
qqnorm(res)
qqline(res, col = "red")
```



Almost all the ressidual data points lay on the normality line.

g)

```r
fc <- forecast(fit, h = 24, level = c(80, 95))
plot(fc)
```


Forecasts from ARIMA(0,1,1)(0,1,1)[12]
6.5
6.0
5.5
5.0
1950
1952
1954
1956
1958
1960
1962